UNIVERSITY OF DERBY
College of Science and Engineering

Timed In-Class Test

OPERATIONAL RESEARCH
5CM522

DATE: MAY 2026
TIME ALLOWED: 4 HOURS

Instructions to Candidates

You are required to answer a total of 4 questions:

However, you can provide answers to all 6 and the best 4 will be considered.

All work must be submitted before the END TIME.

Answers should be given or labelled as per the instructions within the submission template and submitted on Course Resources.

Please submit any accompanying handwritten or typed answers.

You may use Excel or Python, but you must submit all files, and files and answers must be clearly labelled and where necessary, written explanation given.

You may access material on the module and search on the internet to clarify terms or concepts.

You MUST NOT collude with other students, students suspected of collusion may be penalised subject to the academic regulations.

You MUST NOT use any AI/Generative AI such as ChatGPT, Claude, Copilot, BING etc. Any use is considered an academic offence and will mean immediate exclusion from the examination. This includes using built-in AI features in software such as Excel, VS Code and Google Colab.

DO NOT TURN OVER UNTIL INSTRUCTED

1.

Riverside Signs produces two display products each week: Standard panels and Premium panels.

Each Standard panel requires 4 hours in printing and 2 hours in finishing. Each Premium panel requires 2 hours in printing and 3 hours in finishing. Printing has 84 hours available per week and finishing has 60 hours available per week. Data shows that at most 18 Premium panels can be sold per week.

The weekly contribution made by each Standard panel is £42 and by each Premium panel is £36.

(a) Formulate a linear programming model for this problem, clearly defining the decision variables, objective function and constraints. (6 Marks)

(b) Solve the model to determine the weekly production plan that maximises contribution. You may use a graphical method, algebra or Python, but your work must be clear. (6 Marks)

(c) Using your answer from part (b), identify the binding constraints and the unused capacity in each department. State whether the market cap on Premium panels is active. (3 Marks)

(d) Management can obtain extra capacity for one week only, either in printing or in finishing. State which department's extra capacity is more valuable and determine the maximum price per hour Riverside Signs should be willing to pay for the chosen extra capacity. Justify your answer clearly. (6 Marks)

(e) A retailer now requires that at least 30% of all panels produced each week must be Premium panels. Reformulate the model and determine whether the original optimal solution remains feasible or state the revised optimal plan. (4 Marks)

Total 25 Marks

2.

Southvale Council wants to open a minimum-cost set of drone launch sites so that every rural district is covered by at least one site. If a site is opened, it is either fully open or fully closed.

Site	Weekly Cost	Districts Covered
A	16	$\{1, 2, 4\}$
B	11	$\{1, 3\}$
C	15	$\{2, 3, 5\}$
D	14	$\{4, 5, 6\}$
E	13	$\{2, 5, 6\}$
F	10	$\{3, 4, 6\}$

(a) Formulate a model to minimise total weekly cost while ensuring that every district is covered by at least one open site. (8 Marks)

(b) Solve the model in Python and determine which sites should be opened and the minimum weekly cost. (7 Marks)

(c) For your baseline solution, identify any districts that are covered by more than one selected site and comment briefly on the operational value of this redundancy. (2 Marks)

(d) Modify the model so that:

- district 6 must have backup coverage, meaning it must be covered by at least two open sites, and
- sites B and F cannot both be opened because of airspace interference.

Resolve the revised model and comment on the impact of the new restrictions. (5 Marks)

(e) Suggest one realistic way in which this model could be extended to make it more useful for a real emergency-planning application. State clearly what extra data would be needed to support your proposed extension. No full formulation is required, but your answer should be specific. (3 Marks)

Total 25 Marks

3.

PolarAid ships medical kits from three supply depots to four clinics via two cross-docking hubs. Kits may only travel from a supply depot to a hub and then from a hub to a clinic.

Supply Depot	Supply	Cost to H_1	Cost to H_2
P_1	100	4	6
P_2	120	5	3
P_3	80	6	4

Hub	Cost to C_1	Cost to C_2	Cost to C_3	Cost to C_4	Total Demand
H_1	4	6	7	5	
H_2	5	4	3	6	
Demand	70	60	80	50	260

(a) Formulate a transhipment model for this problem. Your answer should define the decision variables, objective function, supply constraints, demand constraints and the flow-balance constraints at the hubs. (8 Marks)

(b) Solve the model in Python and determine the optimal shipment plan, the total transportation cost and any unused supply. Report only the non-zero shipments. (7 Marks)

(c) A disruption occurs and:

- hub H_1 can process at most 130 units, and
- at least 30 units for clinic C_2 must pass through hub H_1 because a specialist inspection step is only available there.

Show clearly how the model must be changed and determine the revised total cost. State only the revised non-zero shipments or the flows that change from part (b), and comment briefly on why the network pattern changes. (6 Marks)

(d) In the disrupted scenario, PolarAid may use an emergency direct route from depot P_3 to clinic C_4 at a cost of £9 per unit, up to a maximum of 20 units. Explain how the model would need to be extended to represent this option and determine whether any direct shipments should be used. State the revised total cost and justify your answer briefly. (4 Marks)

Total 25 Marks

4.

The university IT walk-in desk is modelled as a queueing system. Students arrive randomly, and service times are assumed to follow an exponential distribution.

Use the following assumptions:

Quantity	Value
Simulation length	420 minutes
Average service time	8 minutes
Baseline arrival rate	9 students per hour throughout
Busy-period arrival rate	13 students per hour throughout
Extension service mix	75% standard queries with mean service time 6 minutes, 25% complex queries with mean service time 14 minutes
Primary service target	average waiting time below 8 minutes
Secondary service target	at least 85% of students wait less than 10 minutes
Random seed	42

(a) Describe the main elements of a discrete-event simulation model for this system: entities, resources, events, stochastic inputs and the performance measures you would collect. (5 Marks)

(b) Create a suitable SimPy model to represent the baseline system with 2 technicians and an arrival rate of 9 students per hour. Report the total number of students served, the average wait, the maximum wait and the percentage of students who wait less than 10 minutes. (6 Marks)

(c) Investigate a busier period in which the arrival rate is 13 students per hour throughout the 420 minutes. Test staffing levels of 2, 3 and 4 technicians, present your results in a clearly labelled table and determine the minimum number of technicians needed to meet the primary service target. Comment briefly on whether the secondary target is also met. (4 Marks)

(d) Now consider a more realistic service mix in which 75% of arrivals are standard queries with mean service time 6 minutes, while 25% are complex queries with mean service time 14 minutes. Using the staffing level you recommended in part (c), explain how the SimPy model should be adapted to represent this mix and report the results from one run. State whether your recommendation from part (c) still appears reasonable. (6 Marks)

(e) Explain why a single simulation run is not enough to support a staffing decision and describe how you would use replications and/or confidence intervals to improve the quality of the analysis. (4 Marks)

Total 25 Marks

5.

The operations analyst at Northshore Retail has collected the following quarterly demand data for a fast-growing product line.

Period	Year	Quarter	Demand
1	2019	Q1	380
2	2019	Q2	505
3	2019	Q3	455
4	2019	Q4	590
5	2020	Q1	410
6	2020	Q2	550
7	2020	Q3	500
8	2020	Q4	645
9	2021	Q1	445
10	2021	Q2	600
11	2021	Q3	540
12	2021	Q4	705
13	2022	Q1	490
14	2022	Q2	660
15	2022	Q3	590
16	2022	Q4	775

(a) Using periods 1–12 only, calculate suitable centred 4-quarter moving averages and estimate the trend values where possible. Plot the original data together with the estimated trend. (8 Marks)

(b) Using periods 1–12 only, calculate seasonal indices using both additive and multiplicative decomposition models. (7 Marks)

(c) Use the 2022 data (periods 13–16) as a validation set. Generate forecasts for these four periods using both models and compare their accuracy using mean squared error (MSE), where

$$\text{MSE} = \frac{1}{n}\sum_{t=1}^{n} e_t^2$$

and $e_t = \text{Actual}_t - \text{Forecast}_t$. (6 Marks)

(d) Using the better model, produce forecasts for the four quarters of 2023. Your final recommendation must consider both forecast accuracy and model suitability for this data set, not accuracy alone. Comment briefly on any feature of the data that supports your recommendation, and state whether any single historical observation appears influential enough that you would want to check it before adopting the model operationally. (4 Marks)

Total 25 Marks

6.

Brightmetal Engineering is studying two scheduling problems.

Part A: Flow Shop Scheduling

All five jobs must be processed first on Machine 1 and then on Machine 2.

Job	Machine 1 Time	Machine 2 Time
J_1	8	4
J_2	3	9
J_3	6	5
J_4	7	2
J_5	4	6

Part B: Job Shop Scheduling

Each job has its own machine order and must follow that order exactly.

Job	Sequence on Machines	Processing Times
A	M_1, M_3, M_2	$3, 5, 2$
B	M_2, M_1, M_3	$4, 2, 6$
C	M_3, M_2, M_1	$3, 4, 3$
D	M_1, M_2, M_3	$2, 6, 4$

(a) Use Johnson's Rule to determine an optimal production sequence for the flow shop problem in Part A. Show the main steps of the rule. (6 Marks)

(b) Draw a Gantt chart for your Part A sequence and calculate the minimum makespan. (5 Marks)

(c) Compare the makespan from your Johnson schedule with the makespan obtained from the original order J_1, J_2, J_3, J_4, J_5. Comment briefly on the size of the improvement. (4 Marks)

(d) For the job shop problem in Part B, use a suitable optimisation approach such as OR-Tools to solve the problem. Report the minimum makespan and present a clearly labelled schedule or Gantt chart. Briefly explain how your model enforces:

- the precedence constraints within each job, and
- the machine-capacity or non-overlap constraints.

(6 Marks)

(e) Suppose job D must be completed by time 15 and Machine M_2 is unavailable from time 6 to time 9. State clearly whether the revised problem is feasible. If it is feasible, report the revised makespan and any important schedule changes. If it is infeasible, explain why this is the case. (4 Marks)

Total 25 Marks

End of Paper

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